

Akash Arvind Kore

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SUMMARY

Machine Learning Engineer with 4+ years of engineering experience and 3+ years specializing in **LLM orchestration, NLP, RAG pipelines, and deep learning**. Delivered measurable impact: **90% reduction in ML review time, 35% lower inference costs**, and 99.9% uptime pipelines. Experienced in model evaluation, training, and deployment from research through production on AWS.

EXPERIENCE

AI Software Engineer I

Ready Tensor, Inc.

Aug. 2025 – Present

San Diego, CA (Remote)

- Engineered and shipped **readytensor**, an open-source PyPI tool (Python) for AI/ML repository assessment; built an **LLM evaluation framework** scoring repos across 85 criteria using GPT-4o with seeded, deterministic inference – cutting review time by 90%.
- Unified OpenAI, Gemini, Groq, Anthropic, and OpenRouter into a single **multi-LLM routing layer**, reducing costs by 35%; implemented concurrent processing for batch assessment across a 10-person engineering team.
- Built production **MLOps pipelines** (Docker, AWS Lambda, GitHub Actions) achieving 99.9% uptime; executed zero-downtime database migrations and established CI/CD and IaC standards adopted team-wide.

Machine Learning Intern

Japan Atomic Energy Agency (JAEA)

May 2024 – Aug. 2024

Remote – Raleigh, NC

- Independently designed a **deep learning** pipeline for nuclear reactor bubble detection from scratch; selected **Pix2Pix GAN** in PyTorch for sim-to-real **domain adaptation** under limited labeled-data constraints, enabling reliable sim-to-real generalization for reactor safety.
- Curated a 50k-image labeled dataset and fine-tuned the model via **transfer learning**, achieving **90% object detection accuracy** and 30% improved generalization – eliminating manual labeling dependency.
- Containerized the **real-time inference** pipeline (Docker, AWS Lambda) with ETL preprocessing; achieved 15 FPS throughput and 95% overhead reduction, enabling real-time reactor safety monitoring.

Software Engineer Associate

Siemens Digital Industries Software

Sep. 2021 – Jun. 2023

Pune, India

- Architected Teamcenter on Cloud (TcX) on **AWS (EC2/S3/RDS/Lambda)** supporting distributed data pipelines and telemetry for 500+ global enterprise customers; automated CI/CD via **CloudFormation**, cutting setup time 30%.
- Developed **C++ server-side** components with OOP refactoring, reducing bug resolution time 25%; led cross-functional Python, C++, and Bash debugging across Agile engineering and QA teams, resolving 40+ critical incidents.

Software Developer Intern

Principal Global Services

Jan. 2021 – Sep. 2021

Pune, India

- Built Retirement Income Forecaster (**Django REST API** + React, AWS) achieving 200ms API latency; integrated time-series forecasting models and automated data retrieval via Lambda, reducing generation time 40%.

Research Intern

TATA Research, Development and Design Center

May 2020 – Jul. 2020

Pune, India

- Built an **anomaly detection** library (Python, Scikit-learn, NetworkX) using graph-based ML algorithms (OddBall) on healthcare datasets, improving accuracy by 15%; deployed as a private **PyPI package** adopted by 5+ hospital partners with 98% reliability.
- Built a data preprocessing pipeline (**Pandas, Scikit-learn**) to clean missing values, remove outliers, and balance classes across 10+ clinical datasets; validated with domain experts, enabling reliable deployment across 5+ hospital partners.

PROJECTS

B.I.A.S.E.D. | *FastAPI, Next.js, PostgreSQL + pgvector, Ollama, Docker* | github.com/TheSelfSa/biased

2024 – 2025

- Open-source AI decision intelligence platform; **RAG-backed investigation engine** retrieves evidence from **pgvector** (vector database), routes queries across local Ollama or cloud LLMs, and returns structured responses with citations and confidence traces.
- Built on a **pnpm monorepo** (Next.js PWA + FastAPI backend) with **Better Auth** RBAC, Docker Compose deployment, and a scenario planning engine; released open-source under Apache-2.0.

End-to-End LLM Fine-Tuning Pipeline | *PyTorch, Hugging Face, LoRA, Weights & Biases, Ray*

2025

- Engineered a scalable **model training pipeline** to fine-tune LLaMA-3 using **LoRA/QLoRA** on domain-specific conversational datasets; optimized hyperparameter search with Ray Tune and tracked experiments via **Weights & Biases**.
- Evaluated model performance against baseline using Rouge-L and BERTScore metrics; packaged the final quantized model and deployed inference endpoints via **vLLM** and FastAPI, achieving a 40% reduction in memory footprint.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C++, SQL, Bash

AI/ML: PyTorch, TensorFlow, Deep Learning, NLP, Scikit-Learn, OpenCV, YOLO, TensorRT, ONNX, Hugging Face, Ray

LLMs & GenAI: GPT-4, LLaMA-3, LLM Fine-Tuning (LoRA/QLoRA), RAG, Prompt Engineering, Model Evaluation, vLLM

MLOps & Cloud: MLflow, Weights & Biases, SageMaker, Vertex AI, TorchServe, AWS, Docker, Kubernetes, Terraform, CI/CD

Data & Vectors: Apache Spark, Airflow, Kafka, pgvector, Pinecone, ChromaDB, FAISS, PostgreSQL, MongoDB, Redis

Web & APIs: FastAPI, Django, Next.js, React.js, TailwindCSS, REST APIs, GraphQL, Pydantic

EDUCATION

Master of Science in Computer Science

North Carolina State University

Aug. 2023 – May 2025

Raleigh, NC

Bachelor of Technology in Computer Science

Walchand College of Engineering

Aug. 2017 – Jul. 2021

Sangli, MH, India